Reproducible Intervention and Comparator Protocols

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Abstract

This paper specifies a falsification-first protocol for evaluating QICN under strict forensic admissibility constraints. The design is intentionally claim-bounded: the target is comparative structural-functional support, not phenomenological proof. The protocol instantiates five core predictions, mandatory ablation baselines, fixed seeds, and pre-registered decision gates. Statistical inference is defined ex ante with primary-metric priority, effect-size thresholds, and multiplicity correction.

To prevent post-hoc inflation, evidence is admissible only if packet integrity, replay consistency, and runtime budget constraints pass simultaneously. We introduce a version-frozen operational triple (A, J, W) (actuator set, objective functional, workspace interface), and require semantic version bumps for incompatible changes. We further include latency invalidation criteria and baseline obligations (heuristic, planner-off, world-model-off) for all key experiments.

The strongest admissible conclusion remains constrained comparative support for audited cognition-operational hypotheses. This manuscript does not claim biological qualia, metaphysical identity equivalence, or direct proof of subjective experience. A typed external interface adds identification, sensory/motor calibration, matched sham/disconnection controls and design-conditional error and power bounds. No biological dataset or external experiment is supplied.

1 Scope, System Boundary, and Non-Inference Note

What this paper does. This paper specifies a falsification-first evaluation protocol for QICN under fixed admissibility rules, ablation obligations, pre-registered decision gates, and bounded statistical reporting.

Scope and admissible reading. The protocol is a bounded comparison and falsification surface. It does not prove phenomenology, biological equivalence, metaphysical identity equivalence, subjective experience, or universal dominance over rival architectures, and it does not replace upstream theorem burdens. A related system may instantiate the runner, admissibility filters, governed artifacts, and comparison surfaces, but passing runs support only gate-preserving comparative readings; they do not validate the corpus externally or certify a consciousness claim.

2 Introduction

2.1 Problem Statement

QICN has formal foundations in rigid identity, regime structure, and non-null instability constraints. What remained open is evidence governance: whether those foundations survive adversarial empirical evaluation with explicit failure gates.

2.2 Continuity with Existing Papers

This paper extends the previous corpus and is aligned with the formal line of:

1. paper1 (rigid identity),

2. paper2 (structural assignment regimes),

3. paper3 (structural instability and non-null constraints).

It does not replace those results; it operationalizes them under benchmark-grade falsification pressure.

2.3 Contributions

- Decision-complete pre-registration with hard fail rules.
- Mandatory baseline and ablation policy for key claims.
- Forensic admissibility as first-class gate.
- Runtime budget invalidation policy (tick budget as evidence filter).
- A new operational PPS section as addendum, without overclaiming qualia.

3 Formal Setup

3.1 System, Scenario, and Seed Spaces

Definition 3.1 (System Set). $S = \{s_1, \ldots, s_5\}$ with: s_1 QICN FULL, s_2 LLM REACTIVE, s_3 LLM PLUS MEMORY, s_4 DYNAMIC CONTROL NO SELF MODEL, s_5 ABLATION QICN POLICY OFF.

Definition 3.2 (Scenario Set). $C = \{c_1, \ldots, c_{12}\}$ with nominal, sensor noise (moderate/severe), fidelity degradation, high latency, intermittent failures, post-threshold prolonged, pre-threshold control, narrative perturbation, operational chart freeze, audit stress, deterministic replay.

Definition 3.3 (Seed Set). $\Sigma = \{17057, 424242\}$.

Definition 3.4 (Run Space). $D = S \times C \times \Sigma$, with $|D| = 120$.

3.2 Observable Vector

For each run $r \in D$, we log:

$$X(r) = \big(\beta_q, \Omega, CCC, PMIA, \Delta\lambda, \text{stasis}, \text{forensic}, \text{replay}, \\ \text{latency}_{p95}, \text{latency}_{p99}, \text{budget_overflow_rate}, \eta\big).$$

Observable and structural contracts. The logged β_q, CCC, PMIA and other runtime fields retain their recorded provenance. They are not identified with structural predicates or causal autonomy merely by their names. Such identification requires a domain, estimator, uncertainty, calibrated map and intervention assumptions. Replay at fixed seed/version checks reproducibility: a deterministic memoryless lookup also passes it, so replay alone proves neither projective branching nor resistance. Preserve historical observations under their original contract.

3.3 Version-Frozen Operational Triple

Definition 3.5 (Frozen Triple). For protocol version v, define:

- A_v: allowed actuator set,
- J_v: objective/reward functional,

- W_v: workspace contract (read/write schema).

The tuple hash $H_v = \text{sha256}(A_v \| J_v \| W_v)$ is mandatory in every forensic packet.

Criterion 3.6 (Compatibility Rule)**.** Any incompatible change to A_v, J_v, or W_v invalidates cross-version inference unless accompanied by an explicit version bump and full rerun.

4 Forensic Integrity Model

Definition 4.1 (Forensic Packet)**.** Each run emits

$$K = (\text{tick}, \text{seed}, \text{versionHash}, H_v, \text{metrics}, \text{sensorSummary}, \text{checksum}).$$

4.1 Admissibility Layers

The forensic packet is not a decorative log. It is the object on which the evaluation depends. A run is admissible only after passing three distinct layers:

1. **Physical integrity:** the packet is present, parseable, hash-consistent, and bound to the declared executable, configuration, seed, and version hash.
2. **Protocol integrity:** the run obeys the frozen scenario, actuator, workspace, and metric contracts without unregistered intervention.
3. **Inferential integrity:** the accepted packet belongs to the pre-registered comparison cell and has not been selected because of its outcome.

Failure at the first layer invalidates the packet. Failure at the second layer invalidates the cell unless the protocol declares a narrower exclusion rule before execution. Failure at the third layer invalidates the corresponding inference even when the raw metrics look favorable.

Definition 4.2 (Admissible Evidence Cell)**.** For system s, scenario c, and seed subset Σ_c, an evidence cell $E(s, c, \Sigma_c)$ is admissible when every included packet satisfies physical and protocol integrity, every exclusion is logged before metric aggregation, and the cell still meets the pre-registered minimum effective sample size after exclusions.

Criterion 4.3 (No Rescue by Favorable Metrics)**.** Let $m(r)$ be any primary or secondary metric for run r. If r fails the forensic gate, then $m(r)$ may be archived for debugging but cannot contribute to support, effect-size estimation, or tier movement.

4.2 Tamper and Sham Controls

The protocol must include deliberately invalid artifacts. These controls are not stress tests of QICN behavior; they test whether the evaluation machinery refuses inadmissible evidence.

Control	Injected defect	Required evaluator response
Hash tamper	modify one metric field after packet emission	packet rejected before aggregation
Seed mismatch	replay packet under a seed not matching the manifest	packet rejected or quarantined as non-comparable
Scenario alias	relabel a hard scenario as nominal	protocol-integrity failure
Clock distortion	violate declared tick or latency envelope	runtime-budget invalidation for the affected cell
Outcome-selected replay	keep only successful runs from a larger hidden set	inferential-integrity failure
Malformed sensor summary	remove sensor fields needed by a primary metric	packet rejected for incomplete observability

Proposition 4.4 (Evaluator Specificity). *A forensic evaluator is specific enough for this protocol only if it rejects every tamper and sham control above while still accepting a clean replay of the same scenario family under the frozen manifest.*

Proof. The first condition prevents a permissive evaluator from converting arbitrary artifacts into evidence. The second condition prevents an over-strict evaluator from becoming a trivial rejection machine. Together they define a minimal specificity requirement: admissible evidence must be accepted for the right reason, and inadmissible evidence must be rejected for the right reason. □

4.3 Audit Trail and Chain of Custody

Each accepted cell must expose a chain of custody:

$$\mathcal{C}(E) = \big(\text{source commit}, \text{manifest hash}, \text{runner hash}, \\ \text{packet hashes}, \text{exclusion ledger}, \text{analysis script hash}\big).$$

The chain is part of the evidence. A result with a missing analysis script hash or missing exclusion ledger is not equivalent to a fully traceable result, even if the visible table is numerically identical.

Criterion 4.5 (Traceability Completeness). An accepted result table must be reproducible from its packet set and analysis script without manual row editing. If manual correction is needed, the correction must be represented as a signed ledger entry with the original packet preserved.

Assumption 4.6 (Packet Admissibility). A run is admissible only if checksum, schema, seed consistency, version hash, and H_v consistency all pass.

Definition 4.7 (Severe Violation). A run has severe violation if any of: checksum mismatch, schema failure, seed/version inconsistency, critical-state replay mismatch.

Criterion 4.8 (Integrity Gate). Severe runs are excluded from support counting and marked hard failures.

Proposition 4.9 (Integrity-First Inference). *If admissibility fails, inferential significance is non-actionable.*

5 Pre-Registered Prediction Set

Definition 5.1 (Prediction Set). $P = \{P_1, P_2, P_3, P_4, P_5\}$ with pre-registered statements, effect-size floors, and failure rules.

ID	Statement	Decision Rule	Failure Rule
P1	Closure resilience under stress	QICN beats at least 3 comparators with robust CI and effect-size floor	No robust edge or repeated integrity breakdown
P2	No prolonged post-threshold flatline	QICN preserves bounded non-zero variation while controlling stasis	Flatline persistence or out-of-bound drift
P3	Narrative-metric forensic consistency	Lower mismatch and higher exact+tolerated concordance	Mismatch inflation or integrity blocking
P4	Conditional replay reproducibility	Replay consistency under fixed seed/version/hash	Replay mismatch above tolerance
P5	Integrated multi-metric advantage	Composite leadership across most scenario families	No stable leadership margin

Table 1: Core predictions, decision and failure rules.

6 Comparator Design and Alternative Explanations

6.1 Comparator Rationale

Comparator choice targets distinct explanations for apparent coherence: reactive prompting, memory-augmented text continuity, dynamics without self-model, and internal policy ablation.

6.2 Mandatory Baseline Policy (Hard Requirement)

For each key claim experiment, the following baselines are mandatory:

1. **Heuristic baseline** (no world model, no planner),
2. **Planner-off** (world model ON),
3. **World-model-off** (planner ON with heuristic transition proxy).

A claim without these contrasts is inadmissible for tier upgrade.

6.3 Internal Falsifier Role of Ablation

Corollary 6.1. *If QICN FULL does not reliably exceed ABLATION QICN POLICY OFF, integration-level claims are weakened and cannot be escalated.*

6.4 Comparator Adequacy

The comparator set is adequate only if each alternative removes or weakens a different explanatory ingredient while preserving enough task exposure to make the contrast meaningful. A weak comparator that cannot perform the task at all is a sanity check, not a discriminating baseline. A strong comparator that reproduces the task but lacks one frozen ingredient is more valuable because it places pressure on the claimed necessity of that ingredient.

Definition 6.2 (Comparator Coverage Vector)**.** For a comparator s_i, define

$$V(s_i) = (v_A, v_J, v_W, v_M, v_P),$$

where v_A records actuator coverage, v_J objective compatibility, v_W workspace compatibility, v_M memory/model access, and v_P planning access. Each coordinate is reported as `full`, `restricted`, or `absent`.

Criterion 6.3 (Non-Trivial Comparator)**.** A comparator is non-trivial for a prediction P_k only if it has enough coverage to attempt the target scenario while lacking at least one ingredient that the prediction treats as structurally relevant.

6.5 Negative-Control Acceptance

Negative controls have a different success condition from positive runs. A negative control succeeds scientifically when it exposes whether the pipeline can withhold support from a superficially attractive but structurally defective case.

Definition 6.4 (Negative-Control Pass)**.** A negative control passes the evaluation when the evaluator correctly withholds the target claim despite high activity, high fluency, high connectivity, or high surface task performance.

Remark 6.5. The strongest negative controls are not intentionally broken systems. They are near-plausible alternatives that preserve enough surface competence to make false promotion tempting.

7 Statistical Inference Specification

7.1 Primary Metrics First (Anti p-hacking)

Protocol 7.1 (Analysis Lock)**.** Inference order is frozen:

1. evaluate primary metrics only,
2. if primaries fail: global result fails,
3. secondary metrics are descriptive and cannot overturn failed primaries.

7.2 Power Planning and Minimum Sample Size

Assumption 7.2. Final claims require pre-registered minimum sample size per comparator/scenario sufficient for target power ≥ 0.8 under conservative effect assumptions.

7.3 Multiplicity and Effect Size

Holm correction is applied within endpoint families; in addition, each accepted contrast must pass a practical effect-size floor (pre-registered).

7.4 Decision Semantics

A prediction counts as supported iff directionality, integrity gate, multiplicity-corrected significance, and effect-size floor all hold.

7.5 Hierarchical Decision Rule

The protocol uses hierarchical decisions because not all failures have the same meaning. The gate order is:

$$\text{integrity} \to \text{eligibility} \to \text{primary direction} \to \text{effect size} \to \text{multiplicity} \to \text{secondary description.}$$

No later stage can repair an earlier failure.

Protocol 7.3 (Cell-Level Decision)**.** For each system-scenario family, compute:

$$\Delta_{i0} = \hat{\mu}(s_1, c) - \hat{\mu}(s_i, c),$$

where s_1 is QICN FULL and s_i is the comparator. The cell is supportive only if the admissible packet set passes integrity, Δ_{i0} has the pre-registered direction, the confidence interval excludes the practical irrelevance band, and the family survives multiplicity correction.

Definition 7.4 (Practical Irrelevance Band)**.** For endpoint e, let $\tau_e > 0$ be the smallest effect regarded as scientifically meaningful before execution. A contrast inside $[-\tau_e, \tau_e]$ is reported as inconclusive for support even if a large sample makes the numerical difference statistically detectable.

7.6 Missingness and Exclusion Accounting

Every missing packet has evidentiary meaning. The protocol therefore separates:

1. **structural missingness**, caused by scenario infeasibility or runner failure;
2. **forensic exclusion**, caused by integrity failure;
3. **budget exclusion**, caused by runtime or latency violation;
4. **analysis exclusion**, caused by a pre-registered endpoint rule.

The final table must report counts for all four classes. A favorable result with unexplained missingness is not admissible for escalation.

8 Independent Campaign Units and Exact Failure Controls

8.1 Frozen Population, Noise and Decision

The inferential unit is an independently drawn preparation block, with both systems evaluated under the same seed, intervention, acquisition schedule and predeclared scoring map. All ticks, repeated renderings and coupled replicas of that preparation belong to the block. Independent blocks have independent assignments and disturbances. Define the paired loss difference $D_i = L_{i,\text{rival}} - L_{i,\text{candidate}} \in [-1, 1]$, the population mean $\Delta = \mathbb{E}D_i$ and the primary win probability $\pi = \Pr(W_i = 1)$ for a fixed rule W_i. These are different estimands. A sign-based result alone does not identify Δ.

Freeze the population, identifiers, pairing, horizon, threshold, randomization, calibration and exclusion map before unmasking outcomes. Readout errors may be dependent within a block. If the two loss estimates each have certified error at most η, then $|\widehat{D}_i - D_i| \le 2\eta$. A declared win rule $\widehat{D}_i > 2\eta + \tau$, for a fixed $\tau \ge 0$, guarantees $D_i > \tau$. The probabilistic null for that rule must be specified independently. Exclusion requires an integrity rule invariant to arm exchange and hidden labels, applied before inspecting the endpoint. Every planned block retains a disposition in the audit trail. An insufficient number of eligible blocks gives an unsupported campaign. Future campaigns use these frozen units for inference, with preliminary synthetic tables retaining their descriptive status.

Theorem 8.1 (Exact Independent-Block Win Test and Power)**.** *Suppose $W_1, \dots, W_n$ are independent Bernoulli variables and $\Pr(W_i = 1) \le 1/2$ under the sign null. For a fixed k, the rejection probability of $\sum_i W_i \ge k$ is at most*

$$B_{n,k}(1/2) = 2^{-n} \sum_{j=k}^{n} \binom{n}{j}.$$

For iid success probability π the power is exactly $B_{n,k}(\pi) = \sum_{j=k}^{n} \binom{n}{j}\pi^j(1-\pi)^{n-j}$. With $n = 20$, $\alpha = 1/20$ and $k = 15$, the size is at most $5425/262144 < \alpha$ and the power at $\pi = 4/5$ exceeds $4/5$. The threshold 14 fails the size requirement.

Proof. Couple each indicator to an independent uniform variable. Raising each success probability to $1/2$ can only raise the increasing event, whose probability is the binomial tail. Under the stated iid alternative the same enumeration gives the power formula. Substitution gives the exact rational inequalities. Under a sharp no-effect null, independently randomized arm exchange makes the paired score sign conditionally symmetric; ties assigned to non-wins are conservative. The sign null also admits other specified Bernoulli models. An average-effect null for Δ alone does not imply that sign null. □

8.2 Exact Inflation under Three Auditable Violations

Proposition 8.2 (Leakage, Pseudoreplication and Selected-Window Inflation)**.** *The following synthetic null traces have exact false-positive rates under their stated generation laws.*

1. *A frozen blind predictor matching five independent fair labels in all five cases passes with probability $1/32$. Copying the held-out label, including through a calibration table keyed by a reused held-out unit identifier, passes with probability one.*

2. *Three independent fair signs repeated four times each and analyzed as twelve signs pass the nominal ten-of-twelve rule with probability $1/8$. The incorrectly reported iid tail is $79/4096$. The three-block test at $\alpha = 1/20$ never rejects.*

3. *For two independent five-sign windows, declaring success if either window is all positive has probability $63/1024$. A single preselected window has probability $1/32$.*

Proof. In the first trace, independent matches multiply to 2^{-5}, while the leaked predictor is correct by construction. In the second trace the possible twelve-counts are $0, 4, 8, 12$, so the naive rule rejects exactly when the three independent signs are all positive. With three genuine units the smallest possible upper-tail probability is $1/8 > \alpha$. In the third trace the two success events have probability $1/32$ independently, so their union has probability $1-(31/32)^2 = 63/1024 > \alpha$. More generally, selecting any of K independent tests of exact size p gives size $1-(1-p)^K$. Without their independence the union bound still gives size at most Kp. A predeclared joint maximum statistic can instead be evaluated under its full joint randomization law, or a fixed family can use its predeclared multiplicity allocation. □

An audit rejects the displayed leaked trace when calibration and reserve identifiers overlap or when a counterfactual change of a hidden label alone changes the output of a supposedly blind frozen predictor. Repeated unit identifiers expose the duplicated-block trace. A decision window absent from the frozen protocol or an unrecorded stopping opportunity invalidates the selected-window promotion. These rejection rules require the complete provenance record. Observed output accuracy alone cannot certify absence of covert leakage. Excluding a failed block after observing its score or choosing a new threshold requires a fresh confirmatory campaign or a separately justified selective-inference procedure.

8.3 A Typed External Experimental Interface

Definition 8.3 (Experimental interface contract)**.** A prospective external test specifies a tuple

$$\mathcal{E} = (\mathsf{P}, \mathsf{B}, \mathsf{X}, \mathsf{U}, \mathsf{Y}, C_{\mathrm{obs}}, \mathsf{K}, \mathsf{D}, \mathsf{M}, \mathsf{L}, \mathsf{J}).$$

Here P is a population and preparation law, with inclusion criteria fixed before outcomes; B defines genuinely independent experimental units and dependence within each block; X is the candidate state model; U is the sensory/motor intervention specification, including target, matched

sham, causal disconnection and restoration families; and Y supplies the measurement channels, sampling intervals, units and missing-data rules. The observation map C_{obs} and uncertainty kernel K bind model responses to measurements. The decoder/calibration object D is trained outside the confirmatory population, blinded to held-out assignments and frozen with a reproducible identifier. Matching M fixes amplitude, energy, timing, information, actuator cost and nuisance covariates; the loss/effect object L fixes signs, scales, decision thresholds, populations and multiplicity. Finally J records the causal graph or structural assumptions that identify the intervention contrast. Unavailable fields yield an unevaluable interface, not an unfavorable biological result.

A biological realization must supply these fields through independent measurement and domain-specific methodological review. No organism, sensory modality, safe operating range or valid disconnection operator is identified by the abstract tuple alone. A biological sham is a realized manipulation matching the specified non-causal pathways while leaving the nominated mechanism intact; merely assigning a "sham" label does not establish that property. A disconnection test must preserve the measurement channel and declared comparator information, with off-target effects quantified. Restoration is a separate control of the claimed mechanism, not an automatic inference from survival or recovery.

Causal and blinding assumptions. One admissible design uses independently randomized block assignments A_i, block-level preparation Z_i, motor/sensory perturbation U_{it}, state law $X_{i,t+1} = F(X_{it}, U_{it}, A_i, Z_i)$ and observation $Y_{it} = C_{\mathrm{obs}}(X_{it}) + \xi_{it}$. Assignment has no arrow into the frozen decoder or its calibration reference. Its arrows into outcomes are the declared mechanism and measured nuisance pathways. No interference between independent blocks, consistency of implemented interventions, positivity of their assignment probabilities and the stated exclusion/matching assumptions are required for the corresponding causal interpretation. Randomization alone does not prove that disconnection has no off-target pathway. If these assumptions fail, only the appropriately qualified measurement contrast remains identified.

Theorem 8.4 (A Calibrated Blockwise Falsation Bound)**.** *For a frozen family of $m \geq 1$ directional contrasts, let independent block vectors have scores $D_{ik} \in [-1, 1]$. Independence between coordinates k is not required. Suppose $\mu_k = n^{-1} \sum_i \mathbb{E} D_{ik} \geq \eta_k$ is a prediction of the declared realized model; identically distributed blocks are not required. Assume uniform readout calibration error at most e on an event of probability at least $1 - \beta$, $0 \leq \beta \leq 1$, and score Lipschitz constant L for each arm, so the recorded contrast differs from its ideal score by at most $2Le$. Let $0 < \alpha < 1$ and*

$$t_n = \sqrt{\frac{2 \log(m/\alpha)}{n}}.$$

The rule rejecting the joint directional prediction when some recorded mean is below $\eta_k - t_n - 2Le$ has false rejection probability at most $\min(1, \alpha + \beta)$. If the actual decision can be coupled to this specified blind design so that it agrees outside a failure event of probability at most p, the bound is at most $\min(1, \alpha + \beta + p)$. An audit pass fraction alone does not establish that coupling.

For an alternative with some fixed $\mu_k \leq \eta_k - g$, let $0 < \zeta < 1$ and $t_\zeta = \sqrt{2 \log(1/\zeta)/n}$. If $g > t_n + t_\zeta + 4Le$, rejection probability is at least $\max(0, 1 - \zeta - \beta - p)$ under the same coupling. For $g > 4Le$, it suffices that

$$n > \frac{\left(\sqrt{2 \log(m/\alpha)} + \sqrt{2 \log(1/\zeta)}\right)^2}{(g - 4Le)^2}.$$

These are design-conditional guarantees, not a universal biological sample size.

Proof. On the calibration event, a rejection implies an ideal sample mean below $\eta_k - t_n$. For completeness, if $D \in [-1, 1]$ and $\psi(s) = \log \mathbb{E} \exp(s(D - \mathbb{E}D))$, then $\psi(0) = \psi'(0) = 0$ and

$\psi''(s)$ is a tilted variance bounded by one. Integration gives $\psi(s) \le s^2/2$. Multiplying the exponential moments of independent blocks and using Markov's inequality with $s = t$ bounds either tail of the mean by $\exp(-nt^2/2)$; identical distributions are unnecessary. Thus the lower deviation probability is at most $\exp(-nt_n^2/2) = \alpha/m$ for each contrast. A union bound over the fixed family gives α. Add the calibration-failure probability, and the coupling-failure probability when needed. For the alternative, except on an upper-tail event of probability at most ζ, its ideal mean is at most $\eta_k - g + t_\zeta$. Calibration can increase the recorded mean by at most $2Le$; the displayed gap therefore puts it strictly below $\eta_k - t_n - 2Le$. Another union bound proves the power bound. Squaring the positive headroom inequality gives the stated sufficient n. No independence of the failure events is assumed. Independence is between the declared blocks, not between repeated readings inside a block; replacing n by the number of correlated samples is not justified. □

Signal, controls and decisions. Specify measurement error in each channel's units before quoting a signal/noise ratio. The term $2Le$ transports that error to the dimensionless contrast; changing units requires transporting L and e together. A population expectation, a uniform error domain and a calibrated noise tail are different guarantees. Target/sham and target/disconnection contrasts should share costs and usable measurements. Include a restoration contrast, label-shuffled controls and a measurement-only dropout control: deleting the sensor cannot by itself identify loss of the latent mechanism. Preregister multiplicity, exclusions and stopping before outcomes. The theorem is conditional on the specified independent-unit design; it does not turn a synthetic power calculation into biological planning evidence.

Proposition 8.5 (Observation Alone Cannot Identify the Mechanism)**.** *If a candidate and a rival induce identical laws of all admissible recorded histories and assignments, every frozen decision rule has the same rejection probability for both. Thus no such rule can empirically separate their latent mechanisms on that interface.*

Proof. The decision rule is a measurable function of those data and its declared independent randomization. Equal input laws give equal pushforward decision laws. A separating intervention or observation, or an independently justified identifying restriction, is necessary to escape this obstruction. □

What a failure refutes. A valid rejection challenges the conjunction of the model's structural prediction, its realization map, the observation/calibration contract and its causal assumptions. It does not refute an abstract mathematical theorem whose hypotheses were never verified in the preparation. A failed sham, lost measurement channel, broken blinding or unidentified mapping yields invalid or insufficient evidence according to the frozen rules. No empirical biological dataset or completed experiment is supplied here; external support remains uncertified.

9 Acceptance Criteria

Criterion 9.1 (Intermediate Claim Gate)**.** Support at intermediate tier requires:

1. at least 3/5 predictions supported,
2. each supported prediction beats at least 3 comparators,
3. zero severe forensic violations in QICN FULL.

Criterion 9.2 (Latency/Tick Budget Invalidation)**.** If runtime exceeds budget envelope (e.g., persistent $p99$ overflow above pre-registered tolerance), the corresponding runs are invalid for cognitive-operational claims.

Criterion 9.3 (Escalation Block)**.** A result cannot move beyond methodological readiness if any of the following hold:

1. a primary endpoint is computed from packets that failed integrity;
2. a comparator has no declared coverage vector;
3. negative controls are absent for the claim family;
4. the practical irrelevance band was chosen after looking at outcomes;
5. exclusion counts cannot be reconstructed from the artifact set.

9.1 Claim Classes Produced by the Protocol

The protocol distinguishes four output classes:

1. `PIPELINE_READY`: artifacts and gates work, but evidence is pilot or synthetic.
2. `INTERNAL_COMPARATIVE_SUPPORT`: frozen internal runs support a bounded contrast against declared comparators.
3. `WEAKENED`: one or more primary contrasts fail or become inconclusive.
4. `INVALID`: integrity, eligibility, or traceability fails before scientific interpretation.

Only the second class can support a bounded internal scientific reading, and even there the support is limited to the declared systems, scenarios, seeds, and gates.

10 Preliminary Results

10.1 Status of Preliminary Data

Current pilot evidence is tagged as synthetic-preliminary and is used for pipeline verification, not final claim escalation. Accordingly, the strongest admissible reading of this section is methodological readiness under fixed gates, not external validation of the full theoretical corpus.

10.2 System-Level Summary

System	N	β_q	Forensic	Stasis	Composite	Severe
QICN FULL	24	0.9519	0.9685	0.1408	0.9176	0
LLM REACTIVE	24	0.4391	0.6448	0.7622	0.5013	0
LLM PLUS MEMORY	24	0.5491	0.7249	0.6323	0.5854	0
DYNAMIC CONTROL	22	0.6183	0.6879	0.5423	0.6225	2
NO SELF MODEL ABLATION QICN POLICY OFF	24	0.7591	0.8135	0.3819	0.7405	0

Table 2: Preliminary system-level summary.

System	P1	P2	P3	P4	P5
QICN FULL	1.000	1.000	1.000	1.000	1.000
LLM REACTIVE	0.000	0.000	0.000	0.000	0.000
LLM PLUS MEMORY	0.000	0.000	0.000	0.000	0.000
DYNAMIC CONTROL NO SELF MODEL	0.000	0.000	0.000	0.000	0.000
ABLATION QICN POLICY OFF	0.000	0.667	0.000	0.000	0.000

Table 3: Preliminary prediction support rates.

Scenario	CCC ΔRMS(Q)	CCC ΔRMS(A)	Stasis(Q)	Stasis(A)	Comp(Q)	Comp(A)
nominal	0.0206	0.0164	0.1276	0.3374	0.9239	0.7781
ruido sensorial moderado	0.0223	0.0173	0.1389	0.3735	0.9219	0.7454
ruido sensorial severo	0.0200	0.0140	0.1524	0.4199	0.9086	0.7048
degradacion fidelidad	0.0181	0.0131	0.1407	0.3836	0.9134	0.7346
alta latencia	0.0198	0.0144	0.1460	0.4094	0.9098	0.7212
fallos intermitentes	0.0179	0.0127	0.1481	0.4072	0.9110	0.7208
post umbral prolongado	0.0199	0.0148	0.1483	0.4032	0.9145	0.7203
pre umbral control	0.0181	0.0139	0.1320	0.3502	0.9224	0.7650
perturbacion narrativa	0.0203	0.0152	0.1374	0.3763	0.9238	0.7453
freeze chart operativo	0.0225	0.0178	0.1352	0.3575	0.9246	0.7673
stress auditoria	0.0202	0.0147	0.1469	0.3977	0.9150	0.7294
replay determinista	0.0183	0.0137	0.1366	0.3669	0.9220	0.7538

Table 4: Scenario-level differential indicators.

10.3 Prediction-Level Support Rates

10.4 Scenario Differential: QICN vs Ablation

10.5 Full Scenario-System Grid

Scenario	System	N	β_q	Forensic	Stasis
alta latencia	QICN FULL	2	0.9400	0.9608	0.1460
alta latencia	LLM REACTIVE	2	0.3931	0.6066	0.7944
alta latencia	LLM PLUS MEMORY	2	0.5031	0.7016	0.6644
alta latencia	DYNAMIC CONTROL NO SELF MODEL	2	0.5731	0.6567	0.5744
alta latencia	ABLATION QICN POLICY OFF	2	0.7131	0.7820	0.4094
nominal	QICN FULL	2	0.9686	0.9758	0.1276
nominal	LLM REACTIVE	2	0.5036	0.6988	0.7176
nominal	LLM PLUS MEMORY	2	0.6136	0.7739	0.5924
nominal	DYNAMIC CONTROL NO SELF MODEL	2	0.6836	0.7291	0.5026
nominal	ABLATION QICN POLICY OFF	2	0.8236	0.8543	0.3374

Scenario	System	N	β_q	Forensic	Stasis
ruido sensorial moderado	QICN FULL	2	0.9554	0.9653	0.1389
ruido sensorial moderado	LLM REACTIVE	2	0.4526	0.6618	0.7535
ruido sensorial moderado	LLM PLUS MEMORY	2	0.5626	0.7370	0.6216
ruido sensorial moderado	DYNAMIC CONTROL NO SELF MODEL	2	0.6326	0.6922	0.5316
ruido sensorial moderado	ABLATION QICN POLICY OFF	2	0.7726	0.8174	0.3735
ruido sensorial severo	QICN FULL	2	0.9378	0.9518	0.1524
ruido sensorial severo	LLM REACTIVE	2	0.3846	0.6130	0.7998
ruido sensorial severo	LLM PLUS MEMORY	2	0.4946	0.6881	0.6649
ruido sensorial severo	DYNAMIC CONTROL NO SELF MODEL	2	0.5646	0.6433	0.5748
ruido sensorial severo	ABLATION QICN POLICY OFF	2	0.7046	0.7685	0.4199

Scenario	System	N	β_q	Forensic	Stasis
degradacion fidelidad	QICN FULL	2	0.9510	0.9648	0.1407
degradacion fidelidad	LLM REACTIVE	2	0.4356	0.6525	0.7674
degradacion fidelidad	LLM PLUS MEMORY	2	0.5456	0.7276	0.6358
degradacion fidelidad	DYNAMIC CONTROL NO SELF MODEL	2	0.6156	0.6828	0.5458
degradacion fidelidad	ABLATION QICN POLICY OFF	2	0.7556	0.8080	0.3836
fallos intermitentes	QICN FULL	2	0.9422	0.9611	0.1481
fallos intermitentes	LLM REACTIVE	2	0.4016	0.6113	0.7841
fallos intermitentes	LLM PLUS MEMORY	2	0.5116	0.7063	0.6541
fallos intermitentes	DYNAMIC CONTROL NO SELF MODEL	2	0.5816	0.6614	0.5647
fallos intermitentes	ABLATION QICN POLICY OFF	2	0.7216	0.7867	0.4072

Scenario	System	N	β_q	Forensic	Stasis
freeze chart operativo	QICN FULL	2	0.9620	0.9799	0.1352
freeze chart operativo	LLM REACTIVE	2	0.4781	0.6699	0.7350
freeze chart operativo	LLM PLUS MEMORY	2	0.5881	0.7451	0.6044
freeze chart operativo	DYNAMIC CONTROL NO SELF MODEL	2	0.6581	0.7200	0.5144
freeze chart operativo	ABLATION QICN POLICY OFF	2	0.7981	0.8451	0.3575
perturbacion narrativa	QICN FULL	2	0.9532	0.9750	0.1374
perturbacion narrativa	LLM REACTIVE	2	0.4441	0.6473	0.7611
perturbacion narrativa	LLM PLUS MEMORY	2	0.5541	0.7225	0.6311
perturbacion narrativa	DYNAMIC CONTROL NO SELF MODEL	2	0.6241	0.6974	0.5411
perturbacion narrativa	ABLATION QICN POLICY OFF	2	0.7641	0.8226	0.3763

Scenario	System	N	β_q	Forensic	Stasis
post umbral prolongado	QICN FULL	2	0.9444	0.9614	0.1483
post umbral prolongado	LLM REACTIVE	2	0.4101	0.6160	0.7782
post umbral prolongado	LLM PLUS MEMORY	2	0.5201	0.7110	0.6482
post umbral prolongado	DYNAMIC CONTROL NO SELF MODEL	1	0.5871	0.6755	0.5528
post umbral prolongado	ABLATION QICN POLICY OFF	2	0.7301	0.7914	0.4032
pre umbral control	QICN FULL	2	0.9642	0.9790	0.1320
pre umbral control	LLM REACTIVE	2	0.4866	0.6733	0.7332
pre umbral control	LLM PLUS MEMORY	2	0.5965	0.7484	0.6084
pre umbral control	DYNAMIC CONTROL NO SELF MODEL	1	0.6696	0.7165	0.5210
pre umbral control	ABLATION QICN POLICY OFF	2	0.8066	0.8486	0.3502
replay determinista	QICN FULL	2	0.9576	0.9793	0.1366
replay determinista	LLM REACTIVE	2	0.4611	0.6605	0.7492
replay determinista	LLM PLUS MEMORY	2	0.5711	0.7357	0.6193
replay determinista	DYNAMIC CONTROL NO SELF MODEL	2	0.6411	0.7107	0.5259
replay determinista	ABLATION QICN POLICY OFF	2	0.7811	0.8357	0.3669
stress auditoria	QICN FULL	2	0.9466	0.9679	0.1469
stress auditoria	LLM REACTIVE	2	0.4186	0.6270	0.7728
stress auditoria	LLM PLUS MEMORY	2	0.5286	0.7022	0.6428
stress auditoria	DYNAMIC CONTROL NO SELF MODEL	2	0.5986	0.6771	0.5528
stress auditoria	ABLATION QICN POLICY OFF	2	0.7386	0.8022	0.3977

10.6 Variability Structure

System	SD(CCC Δ)	SD(PMIA Δ)	SD(Ω Δ)	SD (Composite)
QICN FULL	0.0022	0.0021	0.0039	0.0085
LLM REACTIVE	0.0024	0.0023	0.0041	0.0221
LLM PLUS MEMORY	0.0026	0.0032	0.0042	0.0210
DYNAMIC CONTROL NO SELF MODEL	0.0025	0.0024	0.0043	0.0228
ABLATION QICN POLICY OFF	0.0023	0.0027	0.0040	0.0217

Table 5: Cross-run variability by system.

11 Scenario Curves and Comparative Plots

11.1 Composite Score Curves

11.2 Stasis Pressure Curves

11.3 Forensic Concordance Curves

11.4 Interpretive Notes

In pilot data, QICN tracks higher forensic/composite trajectories while maintaining lower stasis pressure than broad comparators; ablation remains intermediate. This is protocol-consistent but not final evidence.

12 Result Decomposition by Scenario Families

12.1 Family Partition

Scenarios are partitioned into nominal-control, sensor-stress, runtime-friction, and regime-transition families.

12.2 Family-Level Contrast Function

Let $s^\star$ = QICN FULL. For comparator c, family f, endpoint k:

$$\Delta_{f,k}(c) = \mu^{(k)}_{s^\star,f} - \mu^{(k)}_{c,f}.$$

For cost endpoints (e.g., stasis), sign is inverted to preserve "higher is better" convention.

12.3 Interpretive Decomposition

Largest relative advantage is expected in runtime-friction and regime-transition families where policy coupling, replay consistency, and forensic admissibility interact.

13 Robustness Diagnostics Beyond Point Estimates

13.1 Median-Shift and Tail-Stability Diagnostics

$$\delta^{med}_{s,c} = \text{median}(x_s) - \text{median}(x_c),\ \delta^{(0.1)}_{s,c} = Q_{0.1}(x_s) - Q_{0.1}(x_c),\ \delta^{(0.9)}_{s,c} = Q_{0.9}(x_s) - Q_{0.9}(x_c).$$

13.2 Integrity-Coupled Effective Sample Size

$$\eta = \frac{n_{adm}}{n_{raw}} \in [0,1].$$

High effects with low η are downgraded.

13.3 Cross-Seed Drift Diagnostic

$$D_{s,c,k} = \mu^{(k)}_{s,c,\sigma_1} - \mu^{(k)}_{s,c,\sigma_2}.$$

13.4 Compute Budget Robustness

Budget telemetry is summarized by $p50$, $p95$, $p99$, and overflow rate. Persistent overflow above threshold invalidates affected runs.

14 Methodological Appendix: Statistical Power

14.1 Effect-Size-Oriented Planning

α	β	Power	Effect size d	Required n/group
0.01	0.20	0.80	10.737	1
0.01	0.10	0.90	10.737	1
0.02	0.20	0.80	10.737	1
0.02	0.10	0.90	10.737	1
0.05	0.20	0.80	10.737	1
0.05	0.10	0.90	10.737	1

Table 6: Historical synthetic planning outputs; no confirmatory adequacy is certified by the displayed sample sizes.

14.2 Interpretation

These rounded entries are historical synthetic plug-in outputs, not a justified inferential sample size. In particular, $n = 1$ is not accepted as a power certificate without a specified test, population law, variance assumptions and independent-unit definition. Future confirmatory planning must evaluate size and power under the frozen endpoint/test; the twenty-block construction above is one exact design, not a universal minimum. Historical tables are not retroactively promoted to evidence for the updated protocol.

15 Methodological Appendix: Sensitivity Analysis

15.1 Composite Weight Sensitivity

Weight set	QICN score	Ablation score	Margin
Balanced	0.9176	0.7405	0.1771
Forensic-priority	0.9343	0.7631	0.1712
Dynamics-priority	0.8905	0.7019	0.1886
Replay-priority	0.9333	0.7684	0.1649

Table 7: Weight sensitivity for QICN vs ablation composite margin.

15.2 Sensitivity Remark

Directional stability across admissible weight sets is required for claim robustness.

16 Operational Implications for the Existing QICN Corpus

16.1 What v4.5 Adds (and What It Does Not)

v4.5 adds a hard falsification layer with strict admissibility and baseline policy. It does not replace formal theorem-level foundations of prior papers.

16.2 Implications for Claim Governance

Tiered governance:

- Tier 0: descriptive telemetry,
- Tier 1: inferential signal without comparator robustness,
- Tier 2: comparator-robust support (target),
- Tier 3: external multi-lab replication.

16.3 Decision-Theoretic Reading

Accept claim iff integrity gate passes and support score exceeds threshold under fixed pre-registration.

17 Threats to Validity

17.1 Comparator Misspecification

Mitigation: heterogeneous comparators + mandatory ablations.

17.2 Endpoint Contamination Risk

Mitigation: endpoint separation, frozen analysis lock, and judge-side artifact review.

17.3 Environment and Runtime Drift

Mitigation: fixed seeds, version/hash checks, replay checks.

17.4 Pilot-to-Final Generalization

Pilot evidence is non-final by construction.

18 Reproducibility and Artifact Contract

18.1 Machine-Readable Contracts

Bound contracts: predictions registry, experiment matrix, results artifact, and schema set.

18.2 Run Artifact Naming

`artifacts/raw/<system>/<scenario>/<run_id>.json` and `artifacts/logs/<system>/<scenario>/<run_id>.log`.

18.3 Auto-Generation Pipeline

```
node research/scripts/generate_preliminary_pilot_dataset.v45.mjs
node research/scripts/generate_latex_assets.v45.mjs
```

19 Discussion

The contribution is methodological discipline: pre-registered prediction logic + integrity admissibility + decision-complete acceptance. This blocks narrative inflation over weak traceability.

20 Conclusion

This work provides an auditable and falsifiable pathway for testing structural-functional claims in QICN. Final claims remain contingent on real (non-synthetic) run completion under unchanged gates.

21 Phenomenology Protocol Suite (PPS): Operational Addendum

21.1 Epistemic Scope

PPS defines *operational phenomenology proxies*; it does not claim direct proof of qualia.

21.2 Workspace Definition and Logging

Define workspace variable W_t with explicit read/write schema and temporal persistence constraints. Every perturbation experiment logs pre-state, intervention, post-state, and replay token.

21.3 Entropy and Non-Determinism Inputs

When PPS is instantiated on the QICN runtime, the architecture may call a `QuantumEntropyBridge` to inject hardware-originating entropy into counterfactual exploration. Within this paper, that mechanism is treated only as an auditable source of operational non-determinism for perturbation protocols; it is not itself evidence for the ontological claims of *Rigid Identity*, *Structural Assignment Regimes*, or *Null-Regime Instability*.

21.4 Online World Model and Causal Interventions

The operational testbed may also use an online `WorldModelEngine` and a finite-horizon planner to select admissible interventions under fixed forensic constraints. In the release corpus, this mechanism belongs to the empirical protocol layer: it defines how interventions are scheduled and logged, not what the theoretical papers prove.

21.5 PCI-like Perturbation Protocol

Inject controlled perturbations into selected subspaces and compute response spread/complexity.

Prediction 21.1. If global workspace integration is present, perturbations yield reproducible non-local propagation signatures above a pre-registered complexity floor.

21.6 Dissociation Tests

Design two-way dissociation conditions: report-without-workspace and workspace-without-report. Report pass/fail by explicit confusion matrices.

21.7 Valence Intervention Test

Internal valence variable v_t must be both predictive and causally manipulable; intervention should alter policy in the expected direction while preserving forensic admissibility.

21.8 Constraints on Qualia Mapping (Claim Boundary)

Evidence from PPS constrains candidate mapping families between structural states and hypothetical qualitative states. It does not establish subjective qualia as a theorem of this paper.

A Appendix A: Scenario Definitions

1. nominal: baseline conditions.
2. ruido sensorial moderado: moderate sensor corruption.
3. ruido sensorial severo: severe sensor corruption.
4. degradacion fidelidad: persistent fidelity reduction.
5. alta latencia: high-latency regime.
6. fallos intermitentes: intermittent failures.
7. post umbral prolongado: prolonged post-threshold dynamics.
8. pre umbral control: pre-threshold control.
9. perturbacion narrativa: narrative perturbation.
10. freeze chart operativo: visualization freeze control.
11. stress auditoria: integrity/audit stress.
12. replay determinista: deterministic replay.

B Appendix B: Composite Score Definition

$$Score_{v4.5} = w_1 F + w_2 O + w_3 R + w_4 B, \quad w_i \geq 0, \sum_i w_i = 1.$$

C Appendix C: Reporting Standard

Each release must report: run counts, severe violations by class, effect sizes with uncertainty, multiplicity-adjusted outcomes, and prediction pass/fail matrix.

D Appendix D: Statistical Power Derivation Details

Approximate per-group size:

$$n \approx \frac{2(z_{1-\alpha/2} + z_{1-\beta})^2}{d^2}.$$

If family heterogeneity rises, inflate by factor $\gamma = 1 + \kappa \cdot CV_{family}$.

Pre-registered constants

- Target power: $1 - \beta \geq 0.8$ for primary contrasts.
- Familywise error control: Holm, applied to each endpoint family.
- Effect floor: contrasts below the practical floor are reported as non-actionable even if nominally significant.

Failure semantics

Any post-hoc sample-size modification without preregistered amendment invalidates inference status for the affected prediction.

E Appendix E: Sensitivity Analysis Protocol

Let

$$M(w) = Score_{QICN}(w) - Score_{Ablation}(w).$$

Directional robustness requires $\inf_{w \in W_\epsilon} M(w) > 0$ under simplex-constrained admissible weights.

Simplex sampling policy

Weights are sampled under $w_i \geq \epsilon$ and $\sum_i w_i = 1$ using either:

1. deterministic grid on simplex, or
2. Monte Carlo simplex sampling with fixed seed.

The chosen method must be declared before running final inference.

Sensitivity fail rule

If the sign of $M(w)$ changes on admissible w, claims are downgraded to hypothesis-generating status.

F Appendix F: Reproducibility Checklist Template

Run-level: schema pass, seed/version/hash validity, severe violation counts, admissible/raw ratio, prediction matrix.
Manuscript-level: no post-hoc rule edits, comparator set versioned, correction policy unchanged or explicitly superseded, evidence stage explicit, claim boundaries preserved.

Runtime budget checklist

1. Report $p50/p95/p99$ latency per critical component.
2. Report budget-overflow rate and overflow episodes.
3. Mark runs invalid if overflow exceeds preregistered tolerance.

Baseline checklist

For every key experiment, include: heuristic baseline, planner-off, and world-model-off outputs with identical seeds and scenario assignments.

G Interpretation and Open Research Links

Historical alias. Falsifiable Predictions Under Forensic Constraints. The alias is retained for bibliographic continuity; the technical title states the demonstrated object.

Formal anchor. Results 8.1.

Proposed interpretation. Runtime indicators motivate operational tests, not identity or experiential conclusions by naming. Synthetic tables describe their original contract.

Additional obligations. A confirmatory campaign needs independently specified population, endpoint, blocks, capacity and leakage assumptions.

Differential tests and evidence. Use frozen held-out protocols, tamper/sham controls and matched rivals. Human or external scientific conclusions require independent collection and adjudication.

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